ODONT 541.1

CARIOLOGY

Susceptibility

Lecture #5

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<u>Office Hours:</u> By appointment

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Quiz – Lectures 3 and 4

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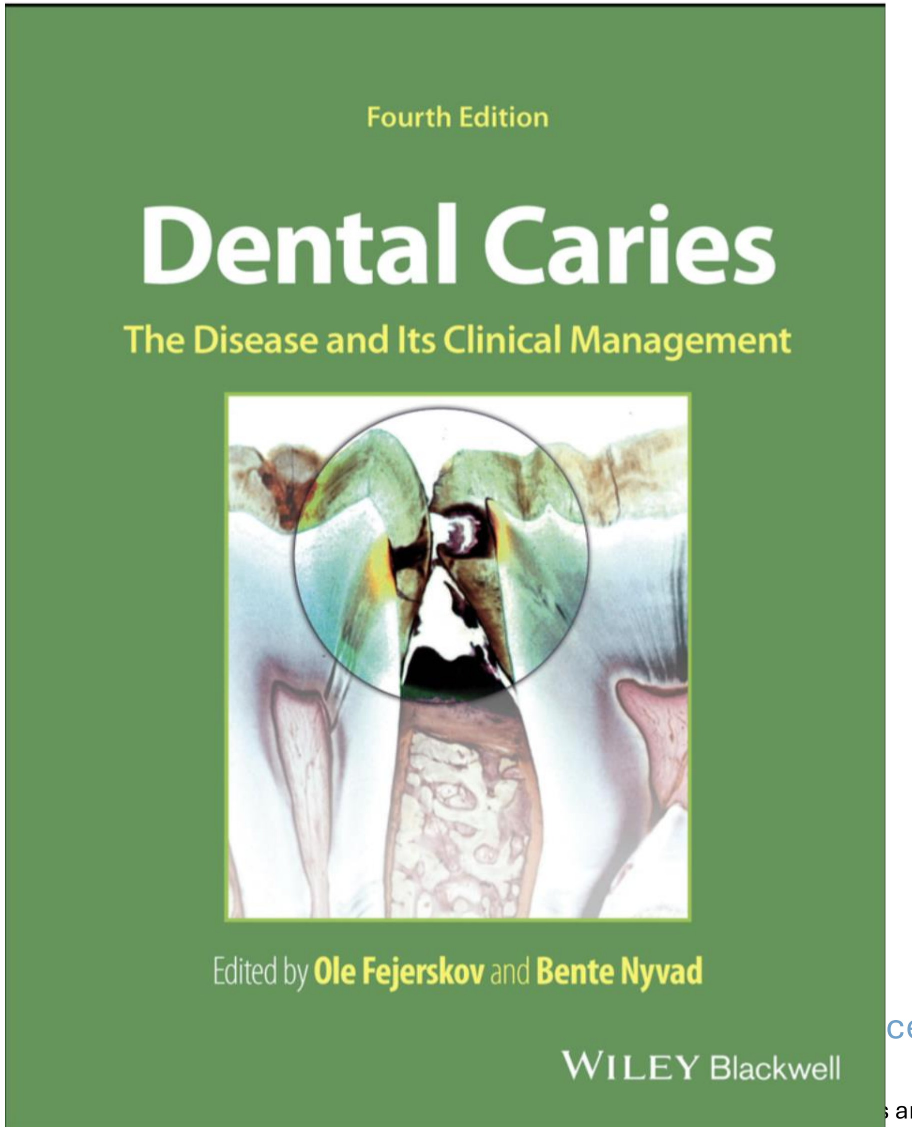
Recommended Reading



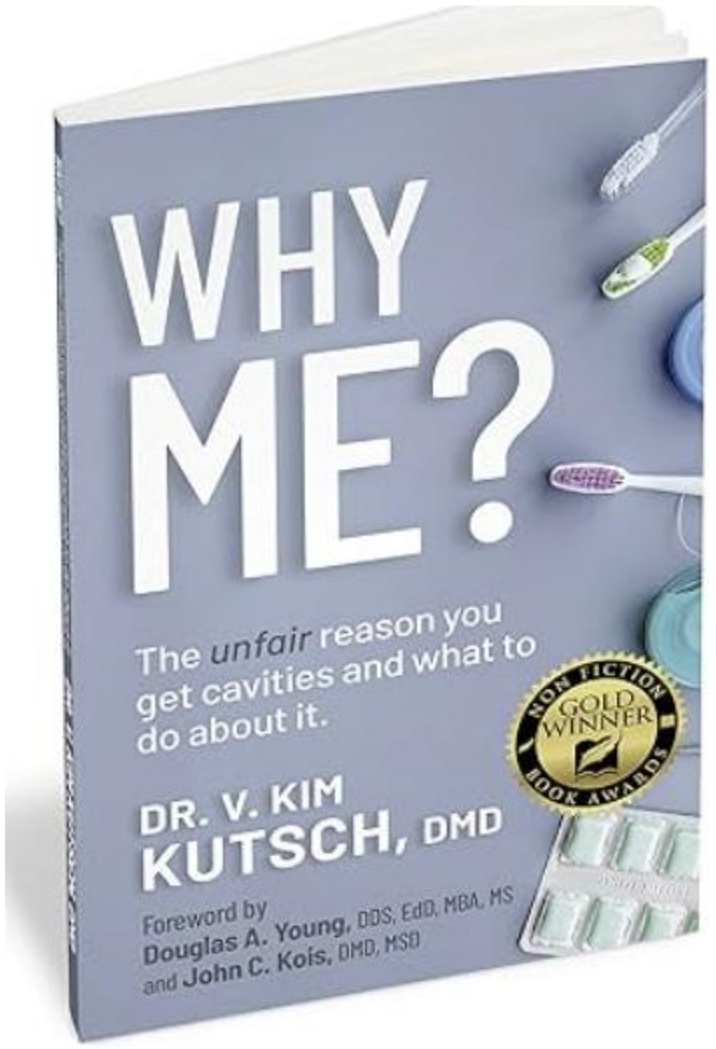
Textbook:

- **Page 153-175 : Saliva**
- **Page 273-279: Oral hygiene**

Etiology

- **Disease management has three levels**
  - **Etiologic control:** remove the cause
  - **Disease control:** stop progression of existing disease
  - **Rehabilitation:** restore lost structure/function

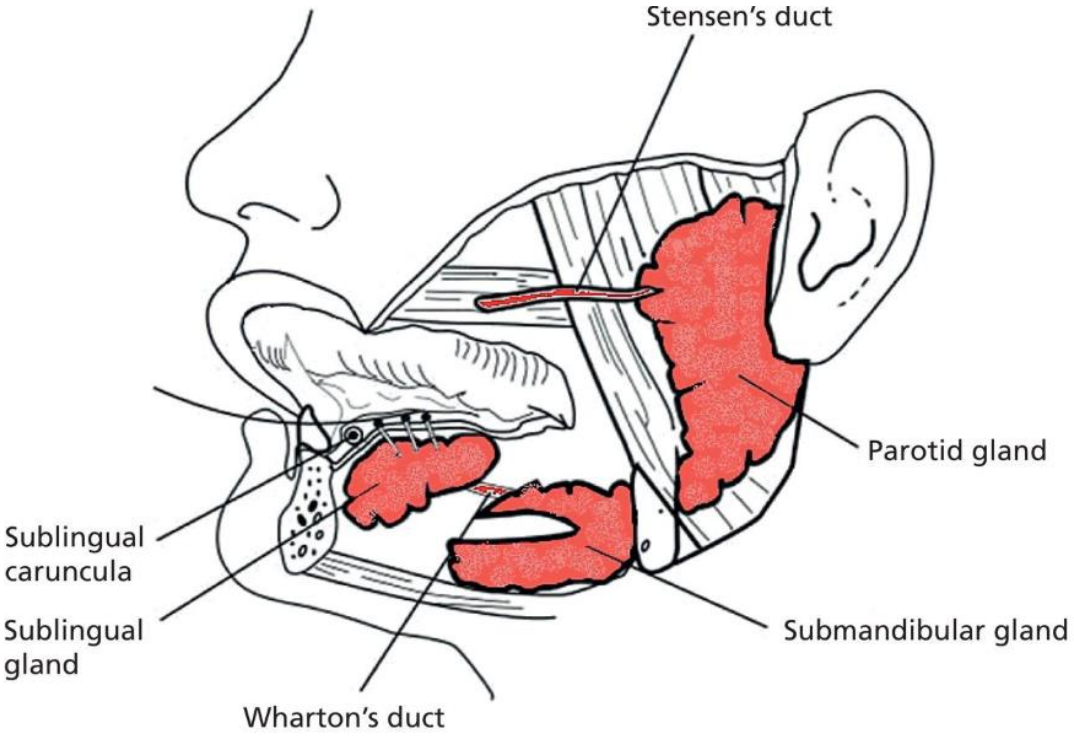
- **Without etiologic control, the disease cycle repeats and compounds indefinitely. Always try to determine WHY this particular patient is prone to caries and what <u>they</u> can do about it.**



More prone to caries		Less prone to caries
Medically Handicapped/ Physically compromised	← Medical History →	No medical or Physical problems
Poor diet - many sugary intakes of food and drink	← Diet →	Good diet - few sugary intakes of food and drink
No fluoride in the water No Fluoride supplements Non- Fluoride toothpaste	← Fluoride →	Water fluoridated Fluoride supplements Fluoride toothpaste
Little ineffective tooth brushing No regular flossing/ cleaning in between teeth	← Plaque Control →	Good regular tooth brushing Good regular flossing/ cleaning in between teeth
Many fillings/ restorations/ missing teeth always new decay and in easy to clean places such as the front of the mouth	← Dental Work →	few fillings/ restorations/missing teeth no new decay
Go only when in pain/ rarely for checkups- very irregular attender	← Dental Recalls →	Go every 6 months or as advised
Little or no saliva (Xerostomia)	← Saliva →	Good normal flow of saliva

Saliva

- More than 90% of total whole saliva is produced by the major salivary glands
  - Composition
    - **Water : 99%**
    - Electrolytes: Sodium, Potassium, **Calcium, Bicarbonate, Phosphate**
    - Organic components: Immunoglobulins, Proteins, Enzymes, Mucins
  - Stimulated by:
    - Olfactory reflex
    - Masticatory reflex (mechanoreceptors)
    - Gustatory reflex (acid chemoreceptors)
  - Resting
    - Diurnal and seasonal
    - Higher flow mid afternoon, **minimal flow during sleep**
      - **Brush before sleeping!**
    - Higher flow in winter than summer



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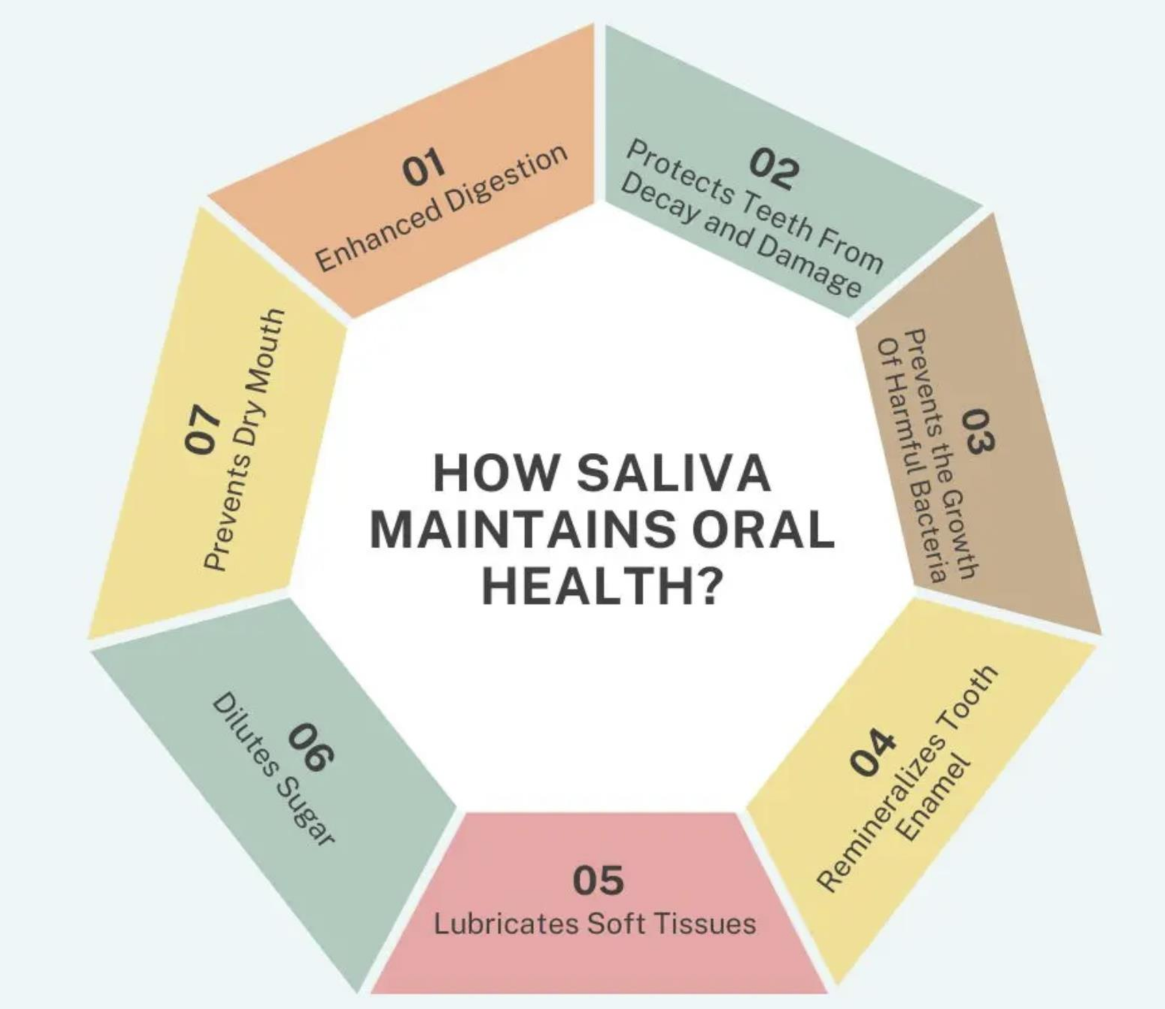
Saliva

Contributions to oral health

- Enhances digestion
  - Moistens and softens foods
  - Breaks down fats and carbs with lipases and amylases
  - Taste perception and activation of downstream digestion
- Protects teeth from decay and damage
  - Buffering
- Prevents growth of harmful bacteria, balancing flora
  - Nutrient support for resident microbes
  - Antimicrobial effects limit overgrowth
- Remineralizes tooth enamel
  - Supersaturation with apatite mineral components
- Lubricating soft tissues for comfort and clearance
  - Also lubricates enamel via acquired pellicle
- Dilutes sugar
  - "The solution to pollution is dilution"



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Saliva

A diagnosis of hyposalivation is established when:

- **Unstimulated whole saliva flow rate is ≤0.1 ml/min**
- **Masticatory (non acidic wax)-stimulated whole saliva flow rate is ≤0.70 ml/min.**

However, xerostomia may occur even in the absence of objective salivary gland hypofunction, suggesting that the feeling of dry mouth may also be attributed to **compositional changes** in the saliva, including impaired **quality** of the residual saliva film.

Salivary gland dysfunction increases risk of:

- Caries
    - Often located at sites generally not susceptible to caries such as the ***cervical margins, incisal margins, or the cusp tips***
- Erosion
- Oral fungal infection
- Frictional symptoms and sores
- Difficulty swallowing and eating
    - Patients may shift towards softer, more cariogenic foods for comfort
- Difficulty with articulation of speech

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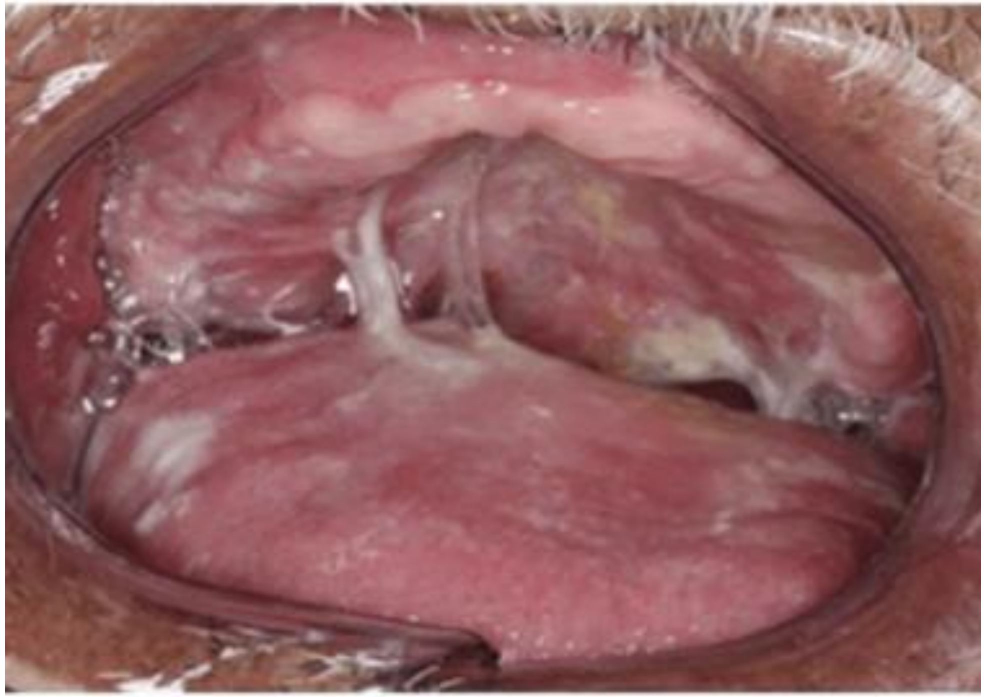

Saliva must FLOW to function.

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Xerostomia is present in approximately:
- 30% of adults over age 65
- 40% of adults over age 80

Both major and minor salivary glands undergo age-related degenerative changes, including loss of acini, increase in fibrosis, and fat tissue.

However, these **structural changes are not necessarily associated with salivary gland hypofunction**, at least not for the function of the parotid glands, which appears functionally unaffected by aging.

Salivary gland hypofunction among older people is more likely to occur due to increased incidence of **systemic diseases** and higher intake of **medication**.

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Table 8.3 Medications and categories of systemic diseases and medical conditions that may cause salivary gland hypofunction and xerostomia [65].

Medications, for example, anticholinergics (like oxybutynin), antidepressants, **anxiolytics, opioids, antihypertensives** **Cannabis
Radiotherapy to the head and neck region
Chronic inflammatory autoimmune tissue diseases, for example, Sjögren's syndrome, rheumatoid arthritis, systemic lupus erythematosus, scleroderma, sarcoidosis, Crohn's disease, ulcerative colitis, autoimmune hepatitis
Endocrine disorders, for example, diabetes mellitus (dysregulated), hyper- and hypothyroidism, Cushing's syndrome, Addison's disease
Neurological disorders, for example, CNS trauma, cerebral palsy, Bell's palsy, Parkinson's disease, Alzheimer's disease, Holmes–Adie syndrome
Mental disorders, for example, anxiety, depression, post-traumatic stress disorder, eating disorders
Infectious diseases, for example, epidemic parotitis, HIV/AIDS, hepatitis C virus, Epstein–Barr virus, bacterial sialadenitis
Genetic disorders, for example, salivary gland aplasia, cystic fibrosis, ectodermal dysplasia, Prader–Willi syndrome
Cancer-related conditions, for example, **consequences of radiotherapy to the head and neck region, orofacial surgery and c**hemotherapy, Graft-vs-Host Disease, palliative treatment
Other conditions, for example, **dehydration, anemia,** burning mouth syndrome, fibromyalgia

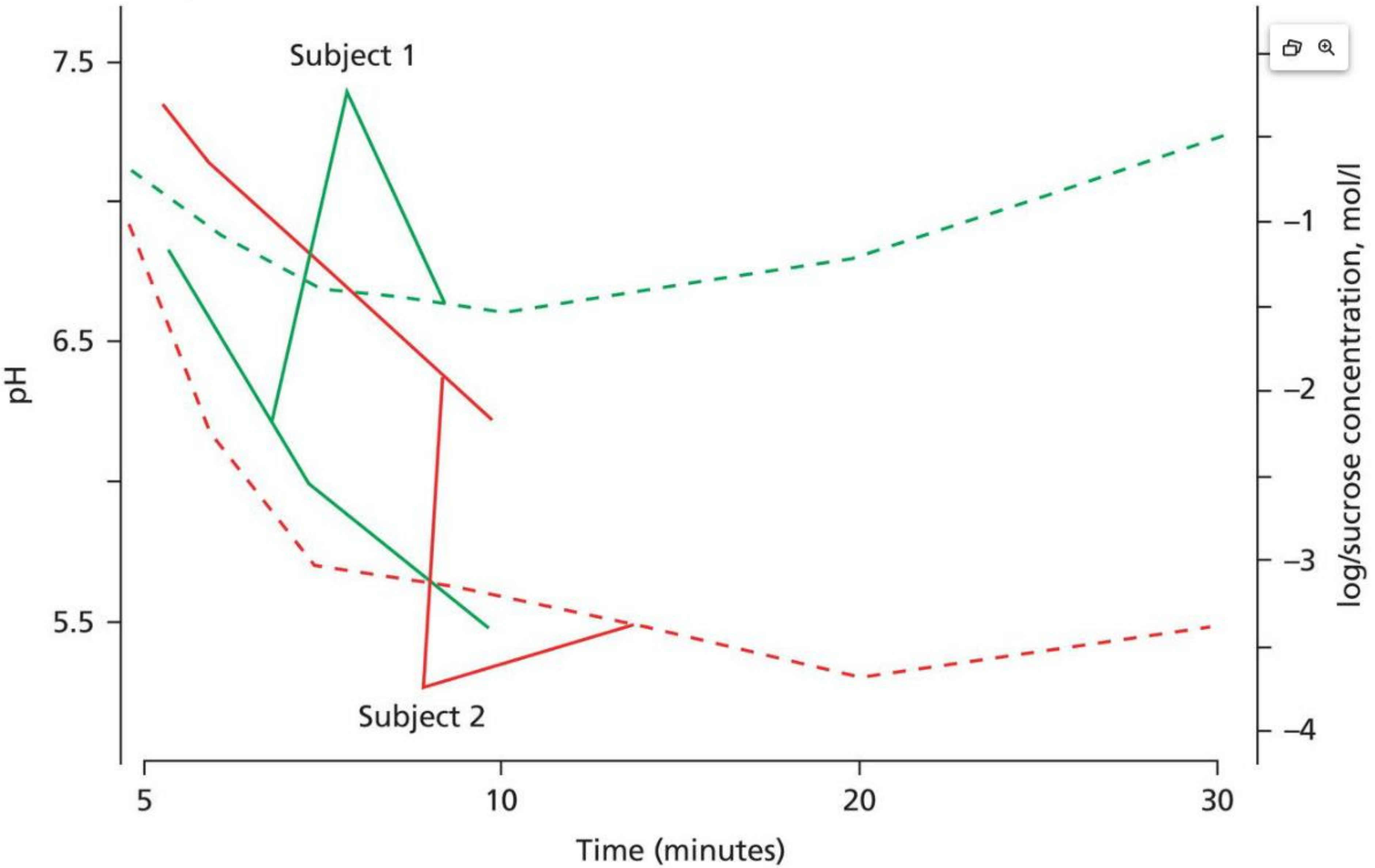
Dietary choices: Excessive intake of caffeine or spicy foods.

Subject 1(green) has normal salivary flow rate. Subject 2 (red) has hyposalivation.

Both subjects rinsed with a sucrose solution.

The solid lines show it took subject 2 **longer to clear the sucrose.**

The dotted lines show the change in plaque pH after sucrose exposure.
Note the **poor/absent pH recovery** in subject 2.



Higher unstimulated salivary flow rate -> shallower demineralized lesions and increased surface mineral content, even in the absence of fluoride or biofilm disruption.

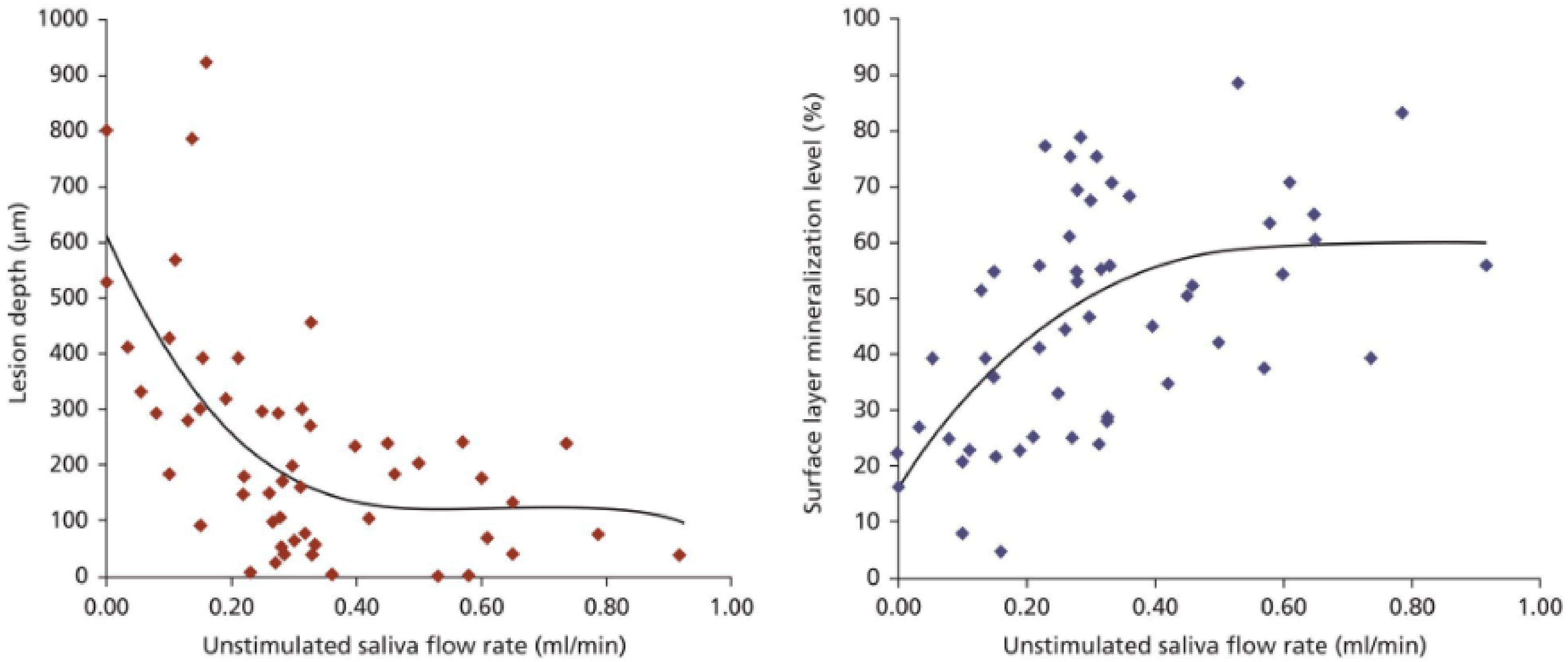


Figure 8.13 The impact of unstimulated whole saliva flow rates on the lesion depth (pm) in experimentally developed root caries lesions and on the mineralization level (percentage of a healthy root surface) in the surface layer of these lesions. Lesions developed during a 2 months period with undisturbed biofilm formation and without exposure to fluoride-containing oral hygiene products [9, 10].

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Table 8.4 Management strategies for xerostomia and the salivary gland hypofunction.

Prevention and treatment
Dental supervision: individual oral hygiene instruction
Regular (every 3–4 months) dental visits and dental plaque control
Regular topical application of fluoride according to individual need
Dietary counselling   Optimize hydration
Prescription medication review (avoidance of xerogenic medication)
Salivary flow stimulation
Physiological: sugar-free chewing gums, candies, and/or lozenges
Pharmacological such as pilocarpine and cevimeline (muscarinic agonists)
Use of salivary substitutes for alleviating xerostomia
Various gels, mouth rinses, and/or mouth sprays
Recognition and treatment of oral candidiasis
Denture hygiene, denture-control, and denture-correction
Antifungals: topical or systemic

Salivary Stimulants

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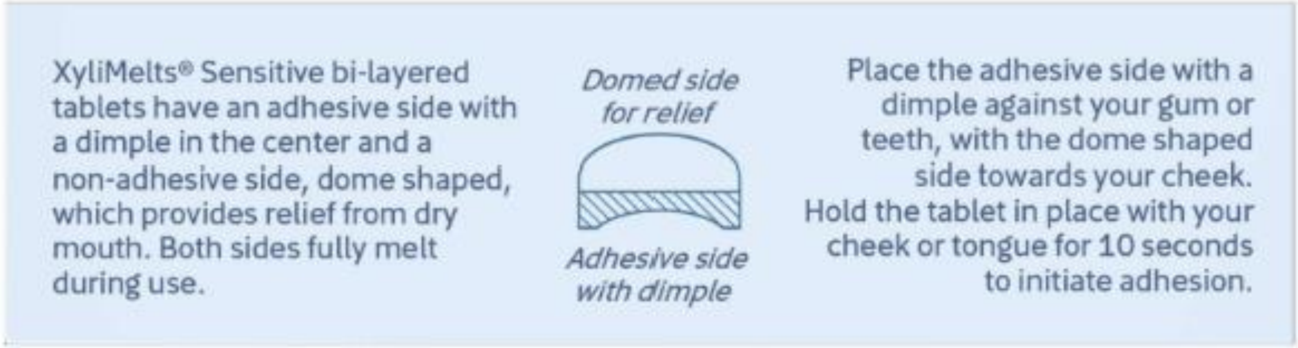

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Salivary Substitutes

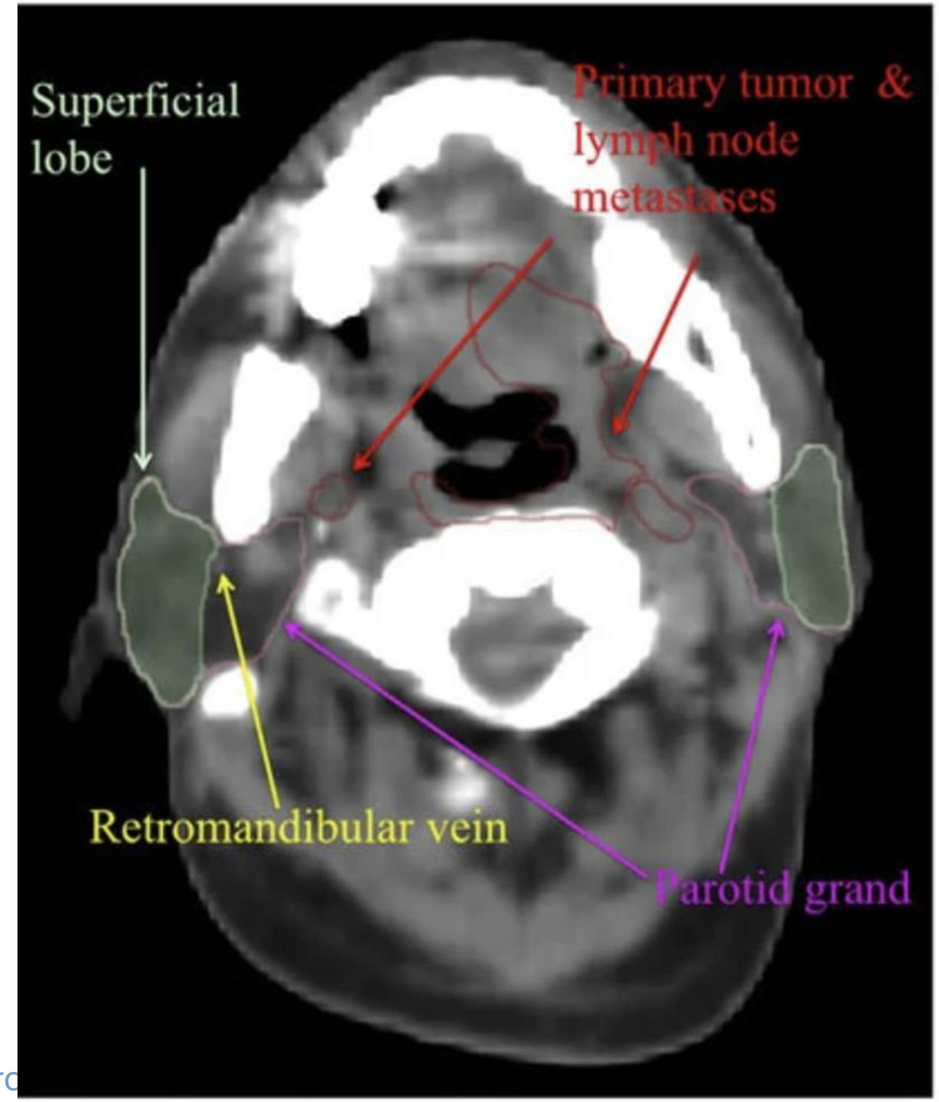
- Biotene (OTC)
    - Lubricant
        - Carboxymethylcellulose and glycerin
    - Enzymatic antimicrobials
        - Lactoperoxidase
    - Anticaries
        - Xylitol
        - Fluoride (some versions)
- Improves subjective dryness symptoms
    - Short duration ~30 minutes
- Does NOT increase natural salivary flow
- Symptomatic relief of mild-moderate xerostomia

Salivary Substitutes

- Neutrasal (Rx)
  - Saliva "reconstitution" product
    - **Supersaturated calcium, phosphate, and bicarbonate system**
    - Buffers pH and supports remin
  - Lubricant
    - CMC and glycerin, like Biotene
  - Use 2-10 times a day
    - Costly, but with potential medical insurance coverage.

Head and Neck Radiation

- Modern standard head and neck radiation protocols are more targeted in an attempt to **partially** spare salivary glands and other unintended anatomy. However, radiation related xerostomia is still common and generally permanent.
- Reduced xerostomia compared to traditional protocols, but will still be xerostomic compared to healthy patients.

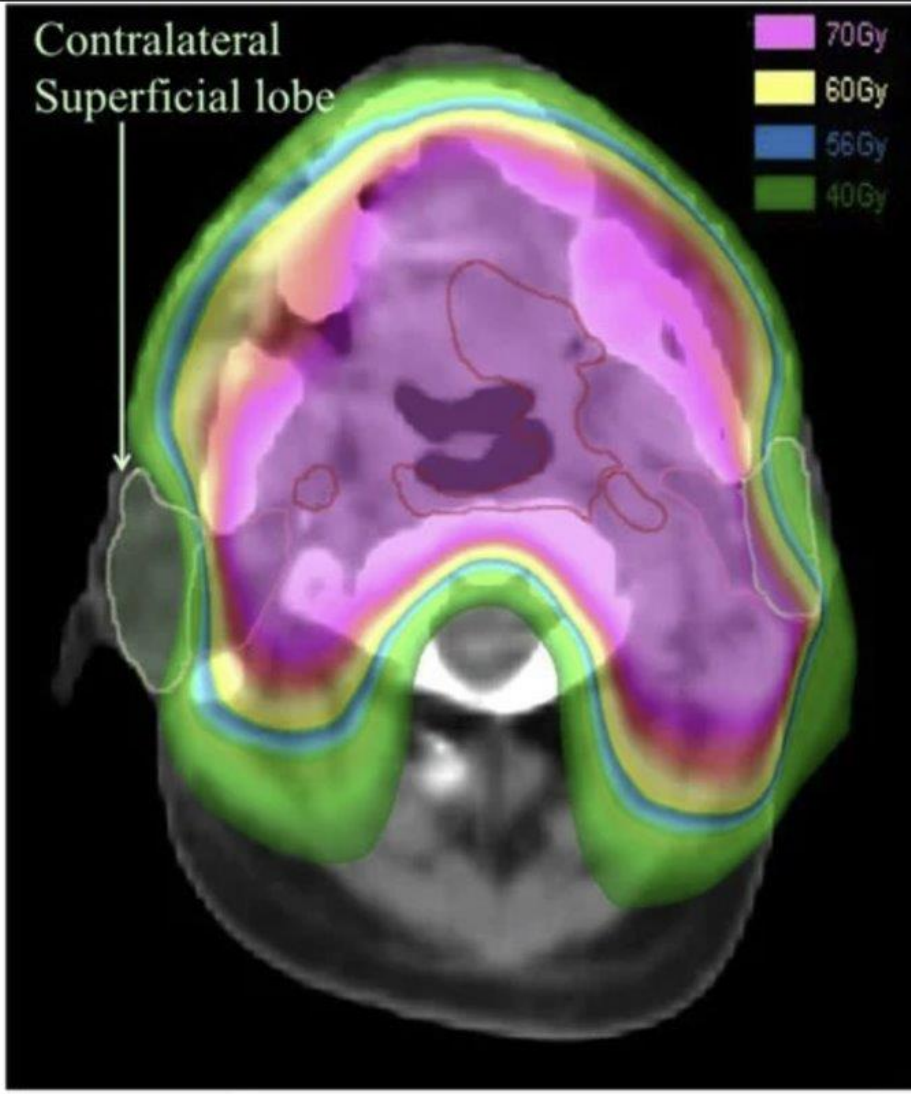


Delineation of parotid gland

Head and Neck Radiation

- Pictured is a modern IMRT (Intensity Modulated Radiation Therapy) approach.
  - More than **26 Grays of radiation to the salivary glands results in clinically significant hypofunction**
  - More than **40 grays results in severe, often permanent xerostomia**
  - Notice that ONE parotid gland is spared. The other received over 40Gy radiation. As did the mandible and teeth.

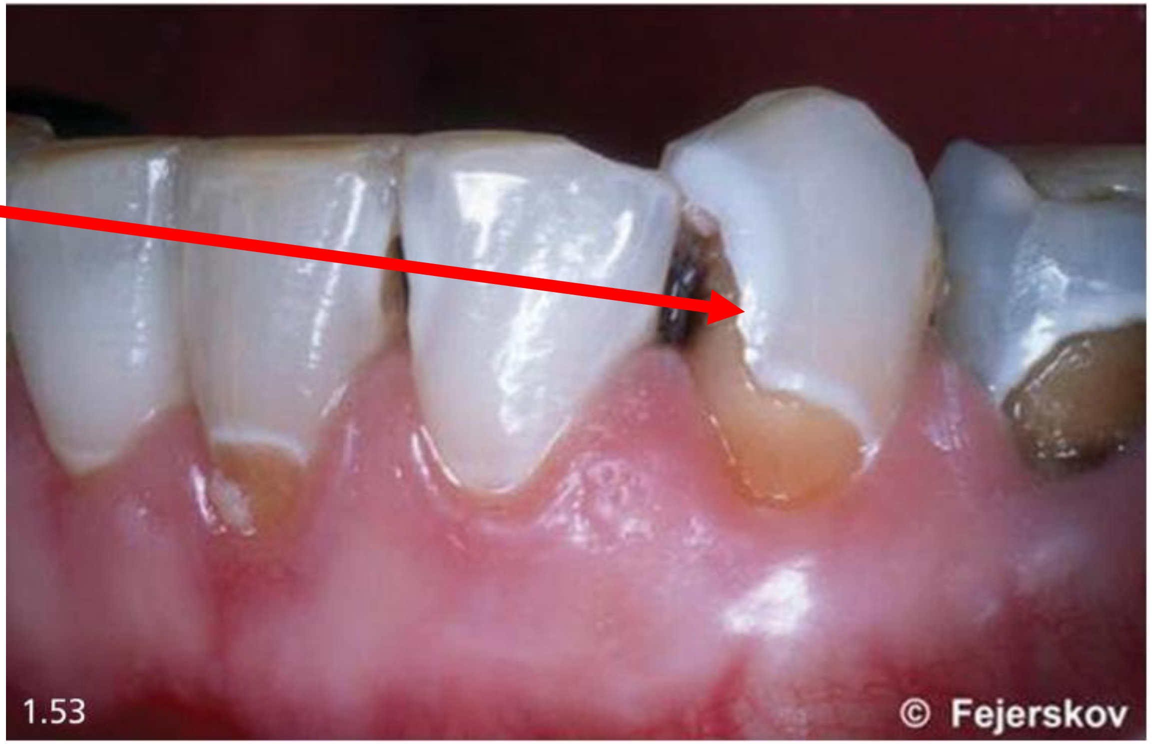
- For patient's scheduled to have H&N radiation in the future
  - Request dose map for path and intensity
  - MD will often recommend xerostomia management products. Collaborate and supplement as appropriate.
  - If possible, prior to radiation complete:
    - Periodontal treatment
    - Caries control
    - Extractions/Root canals
    - Custom fluoridation plan



Dose distribution

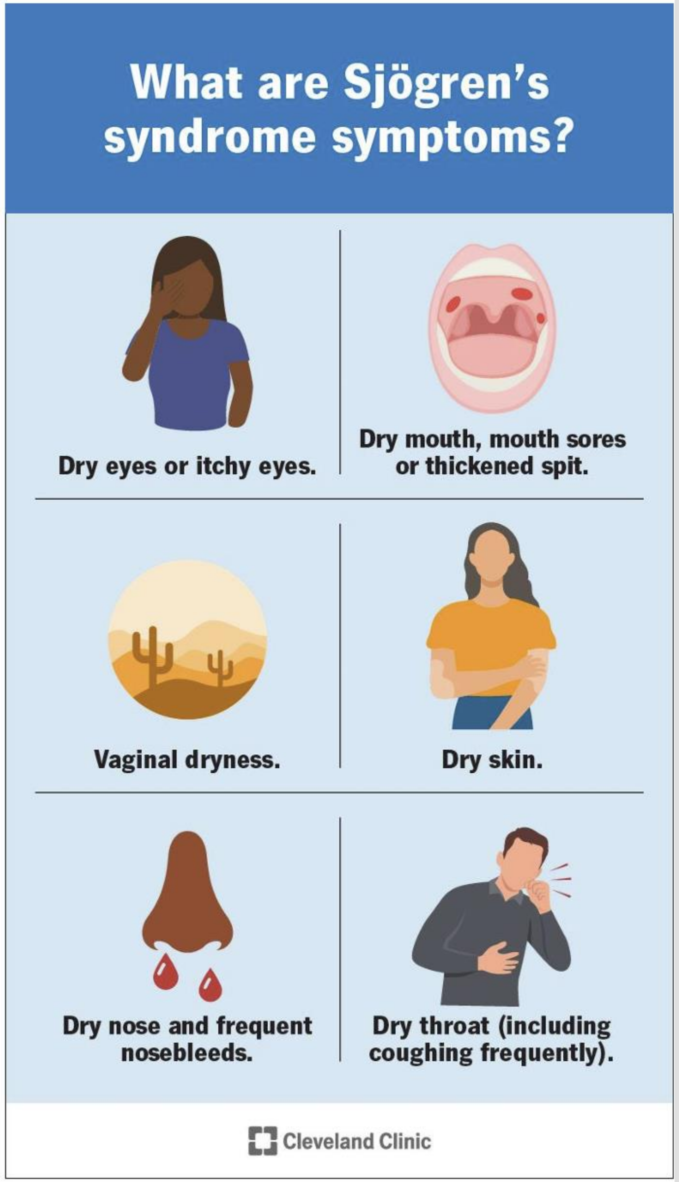
Radiation Caries

- Radiation also directly effects the teeth and bone
- Enamel
  - Alters HA crystal organization, making enamel weaker
  - Increase porosity of enamel
  - Weakens the DEJ bond
    - Higher risk of **enamel delamination or shearing**
- Dentin
  - Radiation effects dentin MORE than enamel
  - Breaks down collagen cross links, damaging the structural scaffolding of intertubular dentin
  - Tubules become wider
    - Teeth more sensitive
    - Caries progresses faster
  - More brittle and less able to absorb stress
  - Odontoblasts damaged -> less able to form sclerotic and tertiary dentin
- **However the DOMINANT cause of radiation caries is XEROSTOMIA**



Sjogren's Syndrome

- Sjogren's syndrome is a chronic **autoimmune** condition that primarily destroys **exocrine glands, such as salivary and lacrimal glands.**
  - Is the second most common autoimmune condition of the connective tissues.
  - Is the systemic condition most frequently associated with salivary hypofunction
  - About 90% of diagnosed patients are women.
  - Chronic inflammation and dysfunction can result in persistent or recurrent swelling of the salivary glands.
- **Many patients with one autoimmune condition have or will develop other autoimmune conditions as well.**
  - YOU may be the first healthcare provider to notice signs suggestive of Sjogren's that may warrant a referral to MD for further exploration and diagnosis.



Alcohol Abuse

- Chronic alcohol -> dehydration -> xerostomia
- Many alcoholic drinks contain cariogenic carbohydrates
- Many alcoholic drinks are acidic
- Alcohol abuse is associated with cariogenic behaviors such as:
  - Poor oral hygiene habits and lack of preventive care
  - Irregular dental visits
  - High risk diets (snacking, sugary foods)
  - Anxiety and depression
    - Xerostomic medications

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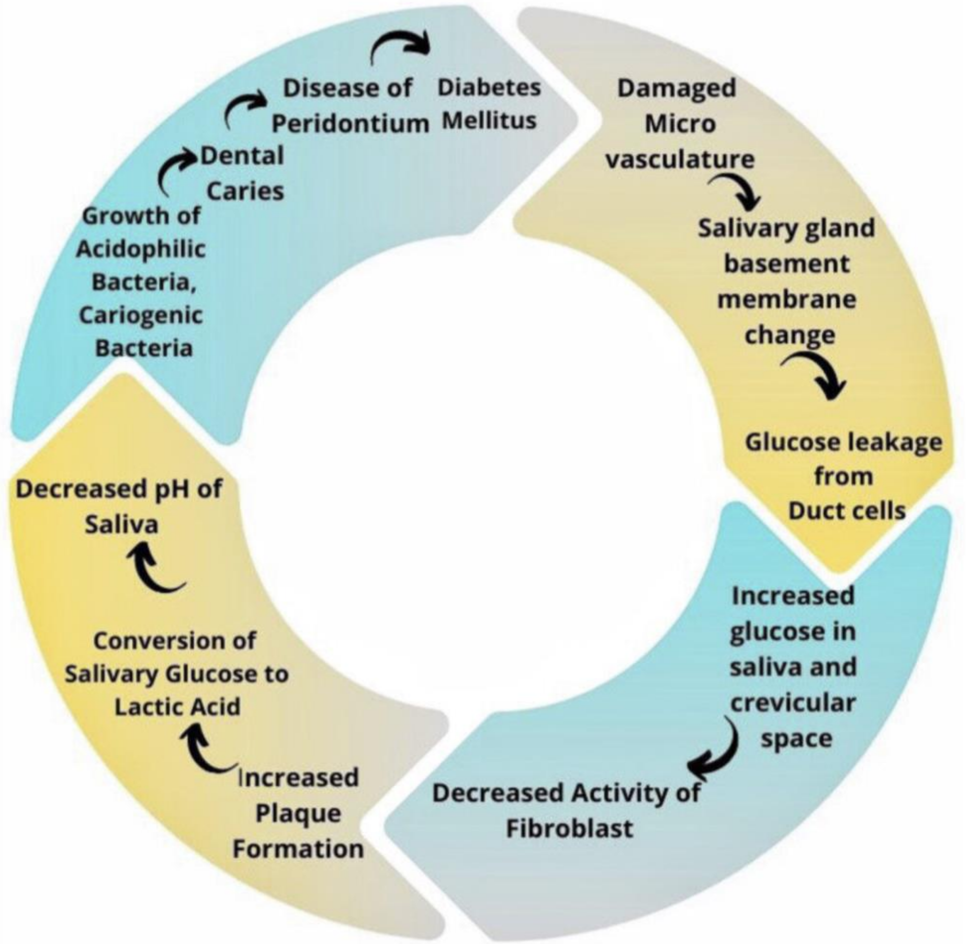
Uncontrolled Diabetes

- Chronic hyperglycemia -> damaged vasculature
  - -> damaged salivary glands -> Xerostomia
  - -> limited delivery of oxygen and neutrophils, specially in smaller capillaries
- Elevated blood sugar -> more glucose diffuses into saliva -> more cariogenic activity in plaque
- Oxidative stress -> damaged neutrophil function -> Impaired/delayed healing
  - Increased incidence and severity of infections, specially secondary infections such as candidiasis
- Increase susceptibility to gingivitis and periodontitis -> root exposure
- Altered dietary habits
  - Increased snacking/frequency of eating to manage glucose swings
  - Increased frequency of drinking to manage thirst



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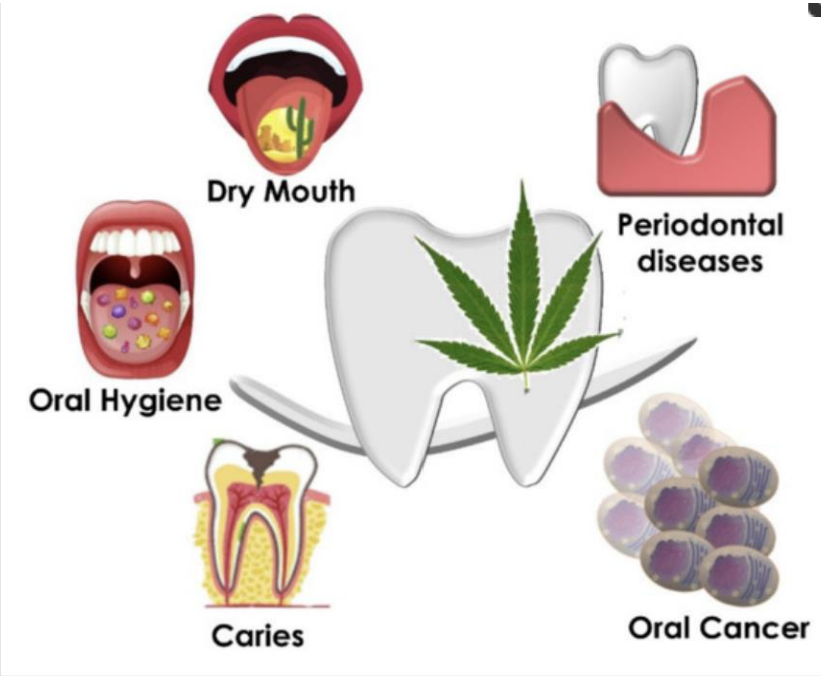
Cigarettes

- Smoking creates a low oxygen environment, favoring anaerobic metabolic activity
  - More S. mutans leads to more EPS production and a **stickier plaque**
    - Nicotine **increases** the activity of lactate dehydrogenase, the enzyme that produces **lactic acid**
  - More lactobacillus, which is highly acidogenic, **sustaining the acidic pH for longer**
- Nicotine is a **vasoconstrictor**
  - Gingival vasoconstriction
    - Increases risk of gingival recession and **root exposure**
    - Decreases GCF production, increasing risk of root caries
    - Gingiva may not bleed, despite gingivitis
- Nicotine is toxic to acinar cells, which produce saliva -> **xerostomia**

Cannabinoids Drugs and Oral Health—From Recreational Side-Effects to Medicinal Purposes: A Systematic Review

Int. J. Mol. Sci. **2021**, 22(15), 8329; https://doi.org/10.3390/ijms22158329

- THC activates cannabinoid receptors which **decrease salivary flow**
- "Munchies" = high frequency of cariogenic carbs
- Associated with poor oral hygiene behaviors
- Also associated with gingival recession -> **root exposure**



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Methamphetamine

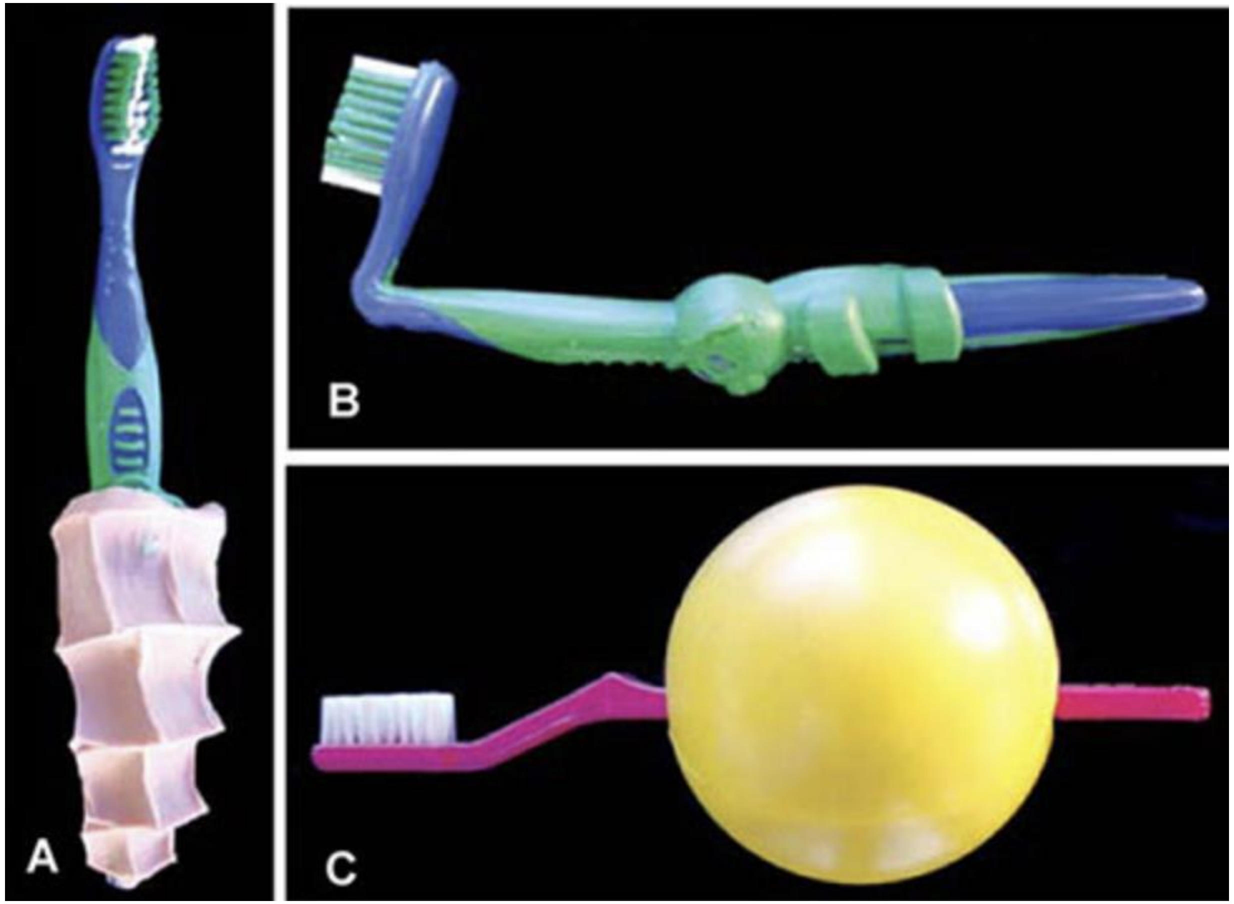
- Methamphetamine itself (when pure) is a weak base, not an acid
- **Intense vasoconstriction and sympathetic stimulation -> SEVERE xerostomia**
- Increased cravings for carbohydrates and carbonated drinks
- Increased bruxism (heavy mechanical forces that can cavitate demineralized enamel)
- Associated with poor oral hygiene
- Maxillary anterior teeth are often first/most affected but caries is **generalized, rampant, and severe**

Mouth Breathing

- Evaporative **localized** xerostomia
  - Commonly presents in **cervical facial surfaces of maxillary anterior teeth**
    - Same surfaces often exhibit localized gingivitis, plaque, WSLs, caries
      - Often mild to moderate
    - Which teeth do most people brush the best?
- Patient's may present with **labial incompetence, open mouth posture, chapped lips, long narrow face**
- Consult with:
  - Orthodontist
  - ENT
    - Nasal obstruction
    - Adenoid hypertrophy in children

Plaque Control

- Hospitalization
    - Dry mouth/intubation
    - Liquid/sugary diet
    - Stress and illness often reduce salivary flow and increase snacking/sugary consumption
- Chronic disability or need for caregivers
    - Parkinson's
    - Alzheimer's
    - Stroke



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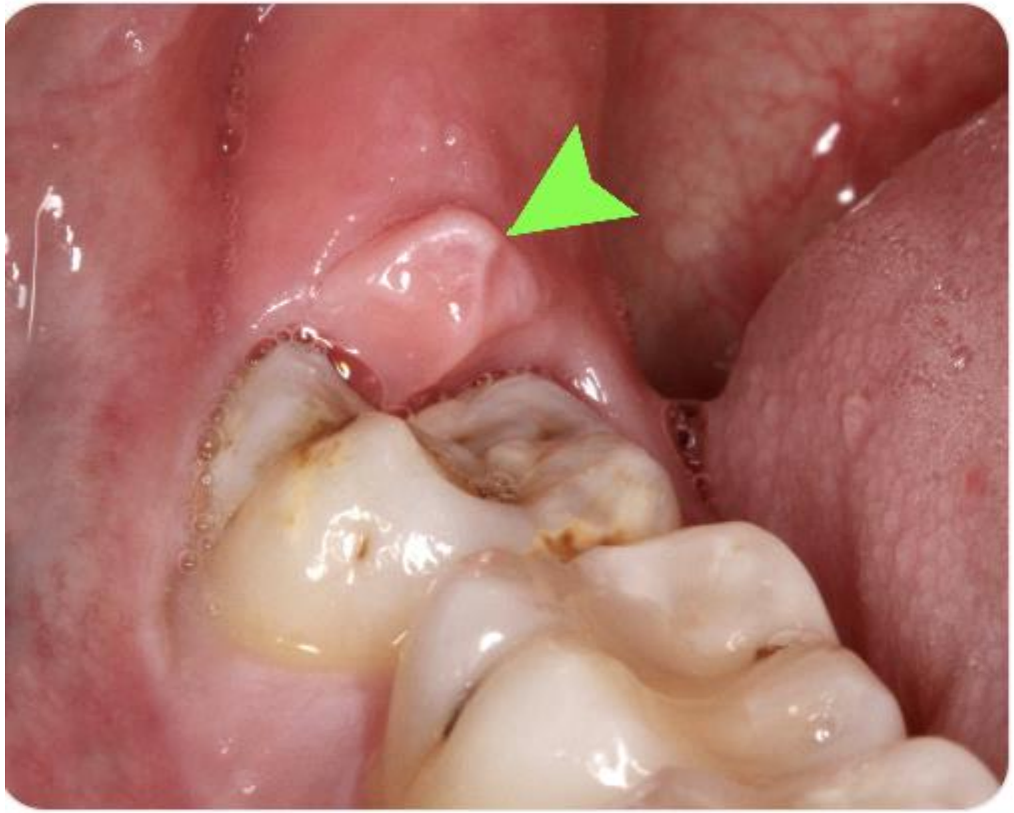
Plaque Control

Partially erupted teeth

- Immature enamel is more susceptible
- Operculum traps plaque

Teeth that erupt earlier have a longer environmental exposure

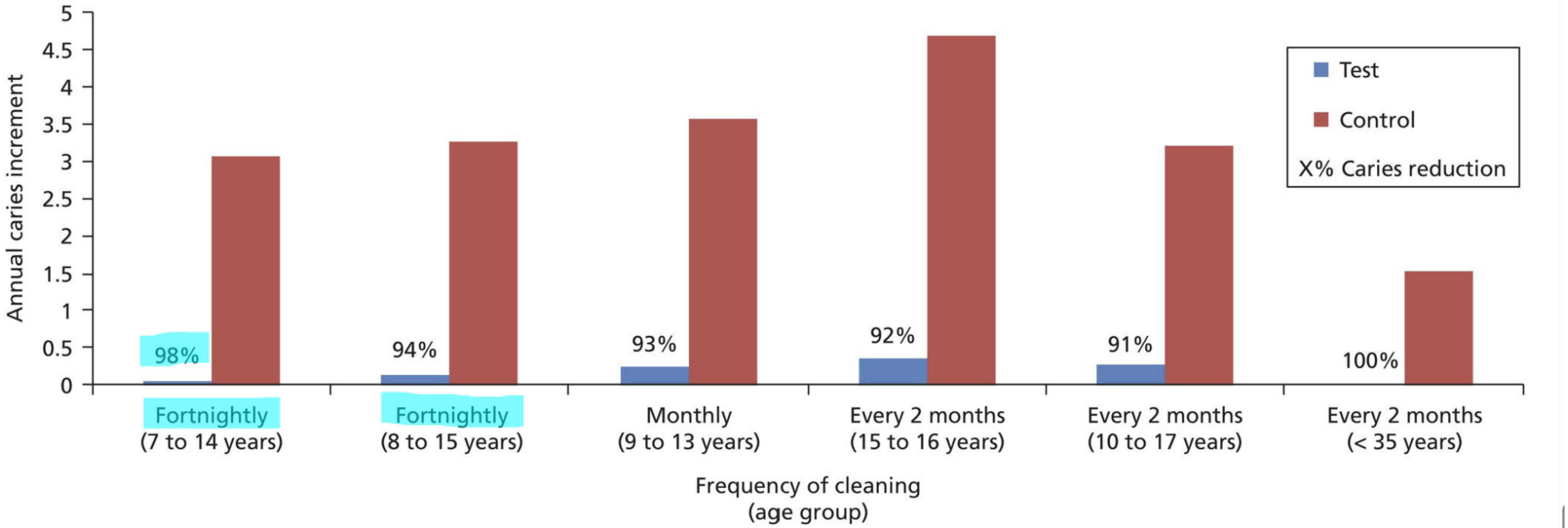
- Permanent first molars



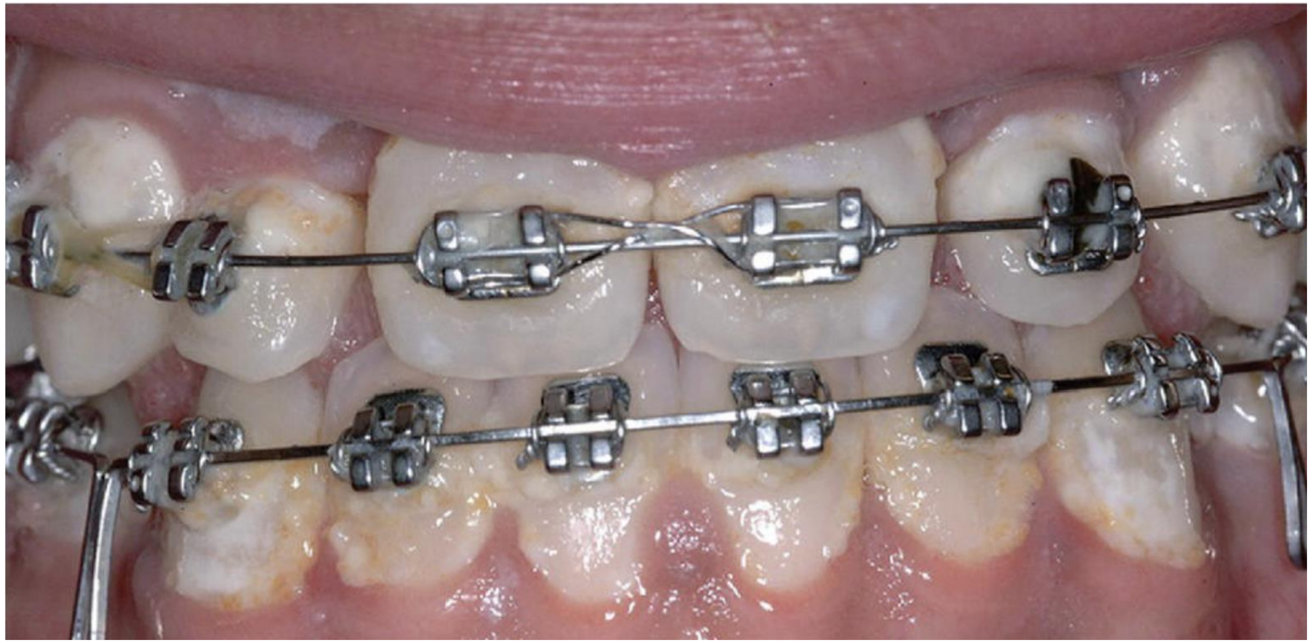
Karlstad Studies

Pg 278 textbook

When plaque is allowed to accumulate on a tooth surface, white spot lesions present within 2-3 weeks.
These children received a professional dental cleaning every 2 weeks throughout the school year. No other intervention or adjustment for confounding factors (genetics, diet, professional fluoride, etc) were made.
Red = no treatment. Average 3-4 caries per year per student
Blue = professional cleaning. Average 0.10 caries per year per students, a 98% reduction.
Disrupting (not eradicating) biofilms at ALL protected sites works!



Special care should be paid to plaque control (both at home and professional) in patients wearing fixed dental appliance that introduce additional protected sites.



Molar Incisor Hypomineralization

Hypomineralization is **not** caries.
However, hypomineralized tooth structure can be more prone to caries and cavitation.

Traumatic Hypomineralization

DOI: 10.1111/ipd.12136 • Corpus ID: 5295180

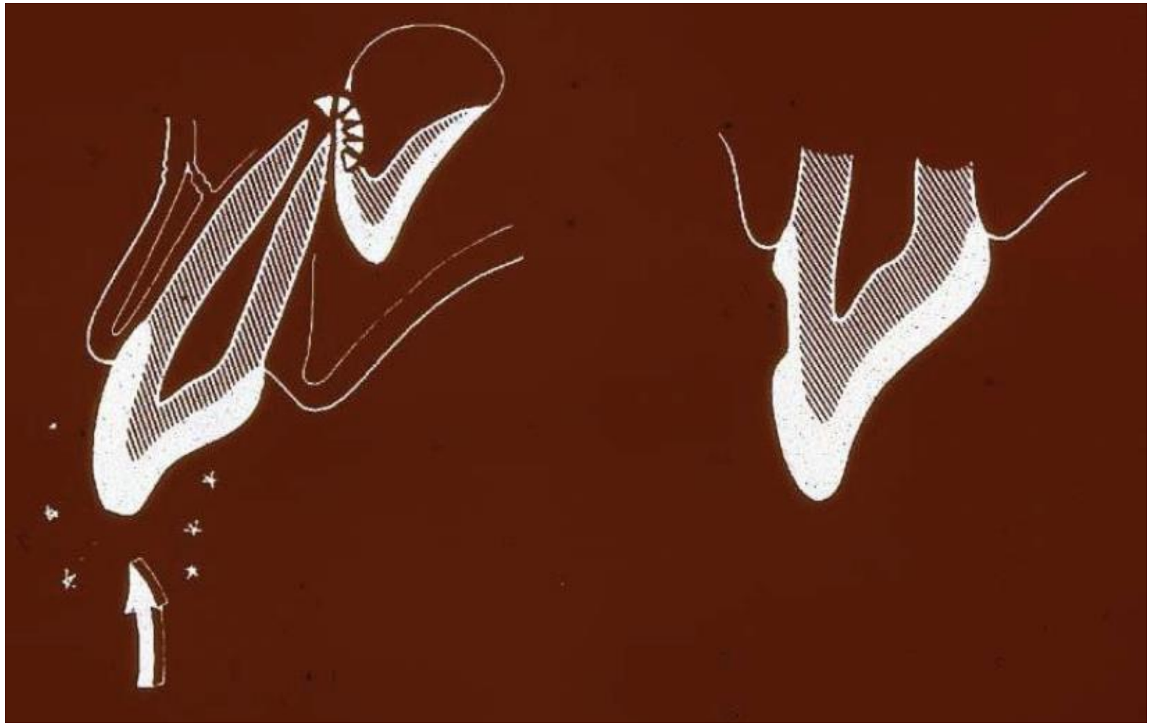
Enamel defects on permanent successors following luxation injuries to primary teeth and carers' experiences.

A. B. Skaare, A. M. Aas, N. Wang • Published in International Journal of… 1 May 2015 • Medicine

Fig. 1. The photo shows teeth 11 and 21 with demarcated opacities >3 mm. Diagnosis: 51 subluxation at 3.5 years. No enamel defects on 6 years molars rule out Molar-Incisor Hypomineralization and the symmetry indicates an idiopathic aetiology according to three independent examiners. Collapse

Trauma to the permanent dental germ

- Dentoalveolar to the primary tooth
- Pulpal infection of the primary tooth
- Iatrogenic during surgical intervention



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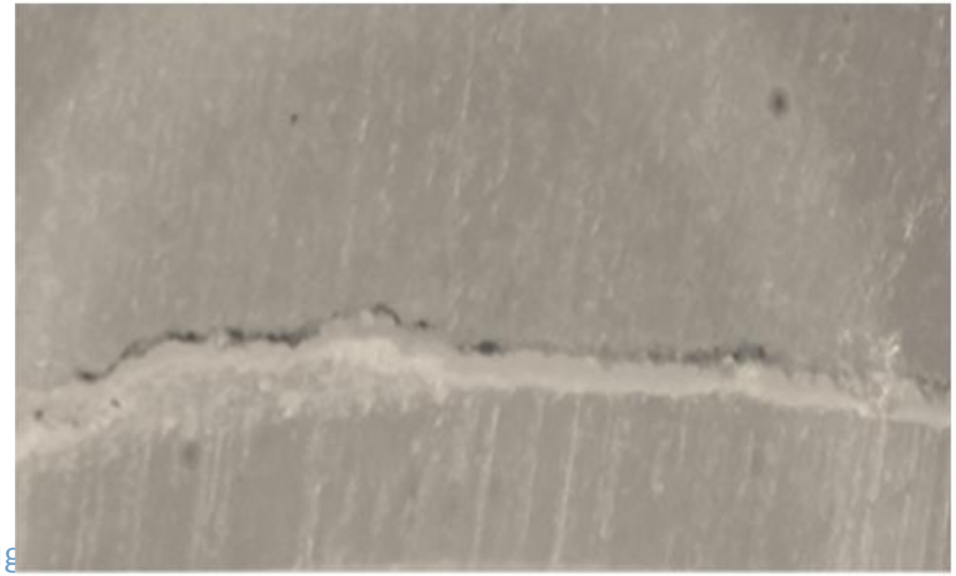
Dental Work

Defective margins (WHEN NOT CLEANSABLE)
- Overhangs
- Open/deteriorated margins
- Contaminated bonds

Recurrent caries: new caries at a restorative margin
Secondary caries: new caries on a tooth that has been previously been treated for caries, but at a site other than the restorative margin

Even in cases of technically ideal restorations, secondary caries will occur if the original etiology is not controlled.



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- The development of a white line at the junction between composite and tooth structure is a mix of broken-down enamel and broken-down composite
  - Because enamel is very weak in tensile strength, these fractures in the enamel could be the result of several factors:
    - Heat from polishing
    - Polymerization shrinkage of the composite or the adhesive
    - Prepping with burs
  - Can also be from the oxygen inhibited layer
- **The white line will eventually turn brown over time and will encourage caries to develop.**

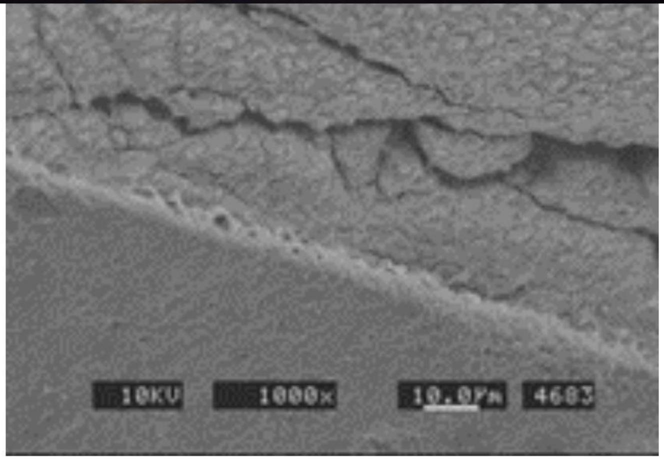

Fig 1. The SEM shows that the adhesive stayed bonded to the enamel interface, but the enamel cracked behind the bonding site. Enamel is very weak in tensile strength; hence polymerization shrinkage can sometimes cause enamel cracks to form.

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